Intro To Database

(Database Fundamental using MySQL)

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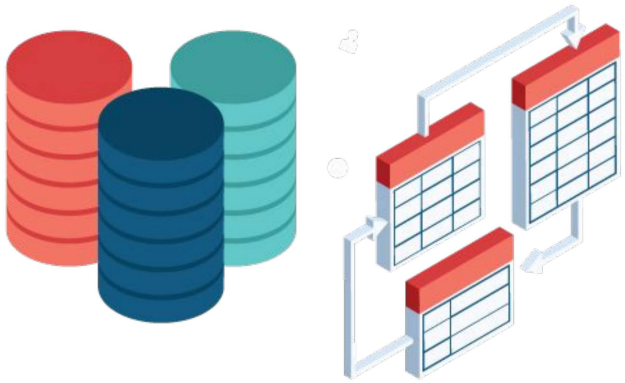
Edited By: Hend Samir

Agenda

- Data & Information.
- File Based System.
- What is Database, Database System?
- DBMS & its functions.
- Database Properties.
- Advantages and Disadvantages of Database Systems.
- DB Architecture.
- Who Deals with Database.
- Data Models.



Data & Information

- **Data** is the raw input (numbers, characters, images…) which when processed or arranged makes meaningful output (**Information**).
- **Data** is the lowest level of knowledge and **information** is the second level.
- **Data** by itself alone is not significant. **Information** is significant by itself.
- Observations and recordings are done to obtain **data**, while analysis and processing are done to obtain **information**.

File Based System

Early attempt to computerize the manual filling system , **each Program defines and manages its own data.**

- Separation & Isolation Of data (each user has a copy)
- Data Redundancy (Duplication of data)
- No Database integrity
- All programs maintain metadata for each file they use
- Incompatible File Formats
- Non-standard way to query

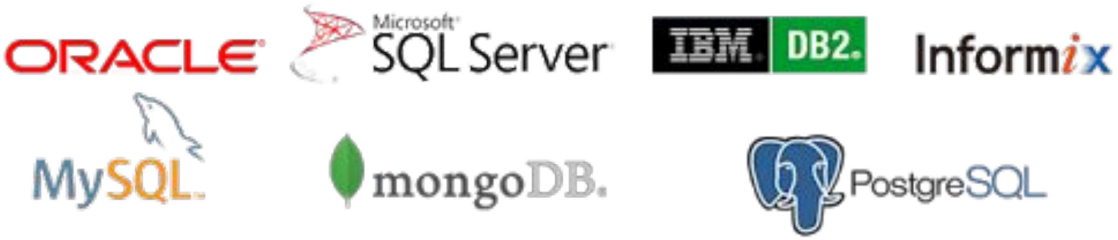
What is a database?

- A database is an organized collection of related data.”

- The data is typically organized to model relevant aspects of reality in a way that supports processes requiring this information

Database Management System(DBMS)

- It is the intermediate layer between the database and the programs that access the data.
- It is collection of programs that enables users to create and maintain a database.



DBMS Functions

- Constructing Database.
- Manipulating Database.
- Data Security
- Data Integrity.
- Concurrency.
- Backup & Recovery.
- Data Dictionary (Meta Data).
- Performance.

DBMS Other Functions

- Text/Number/Image/Audio/Video.
- Special Data (like GIS).
- Time Series(data stored versus the time).
- Data Mining

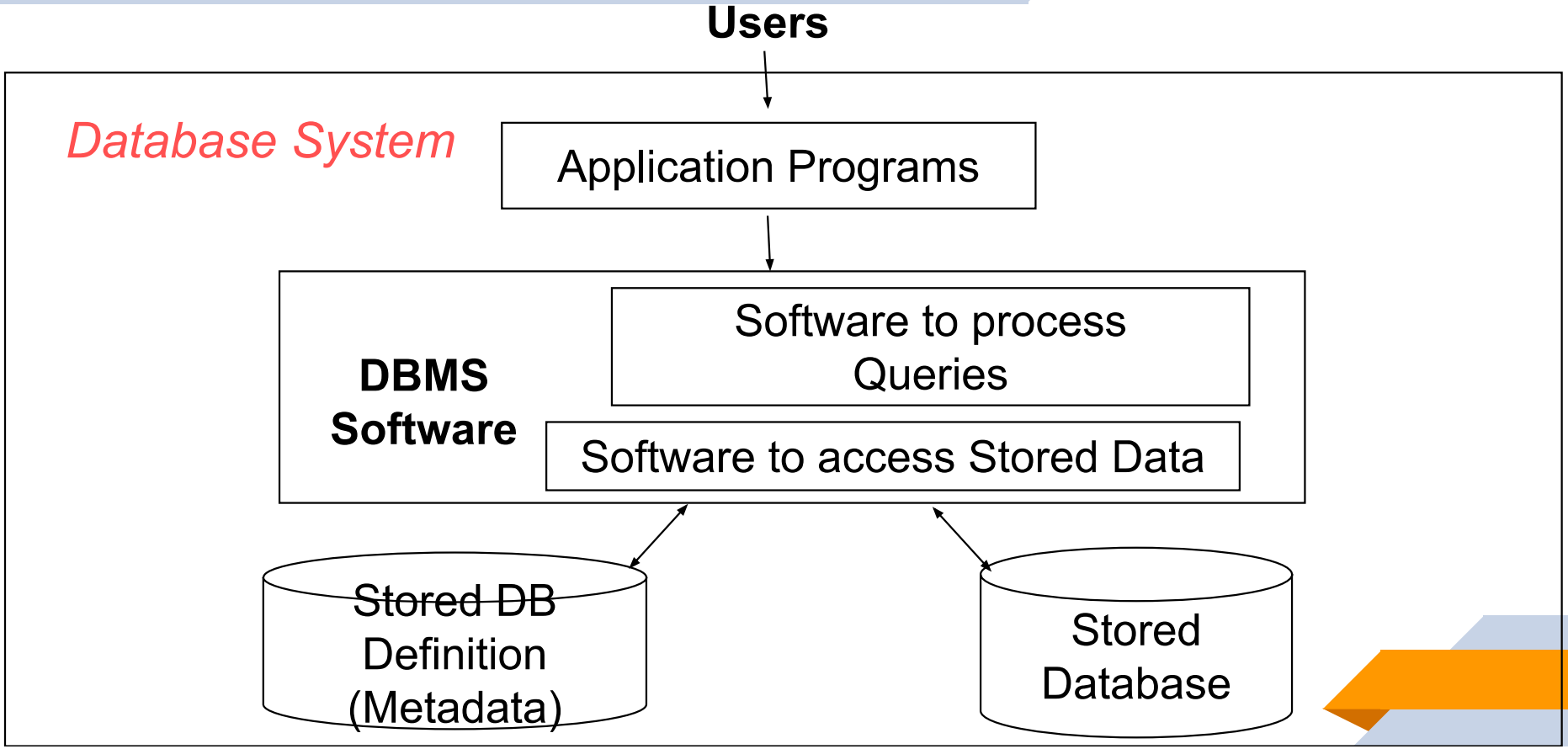
Advantages of Database

- Controlling Redundancy.
- Sharing Data.
- Restricting unauthorized Access.
- Enforcing Integrity Constraints.
- Inconsistency can be avoided.
- Providing Backup and Recovery

Disadvantages of Database

- It needs expertise to use.
- DBMS itself is expensive.
- DBMS may be incompatible with any other available DBMS.

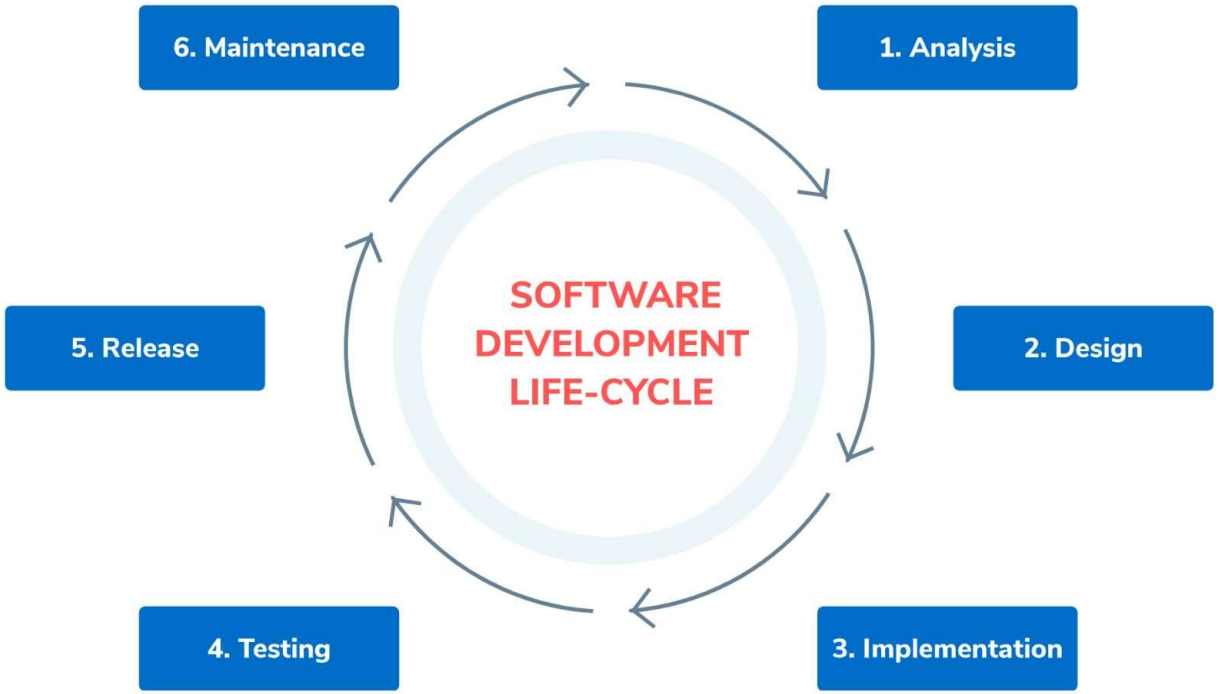
Database System



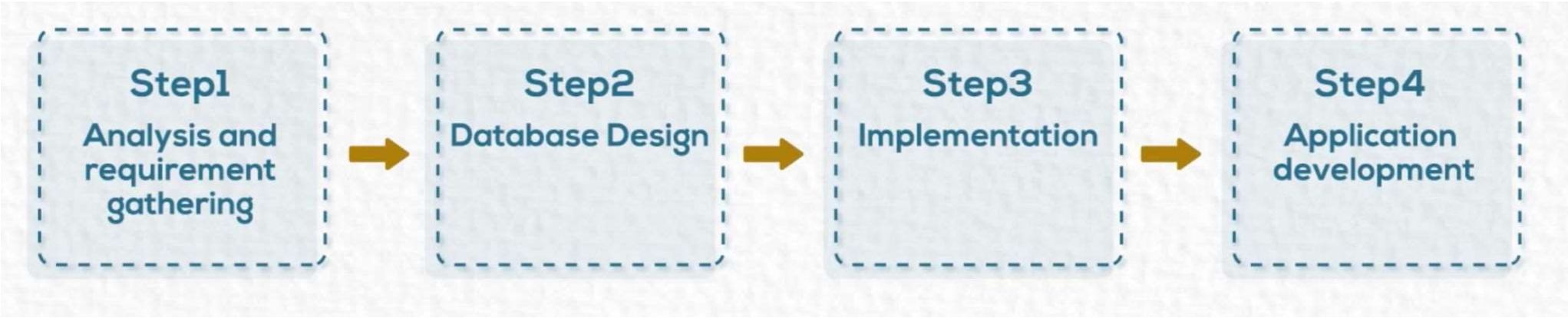
Database System

- A database system is composed of :
  - The database.
  - The Software.
    - Application Program
    - DBMS

Software development life cycle



Database development life cycle

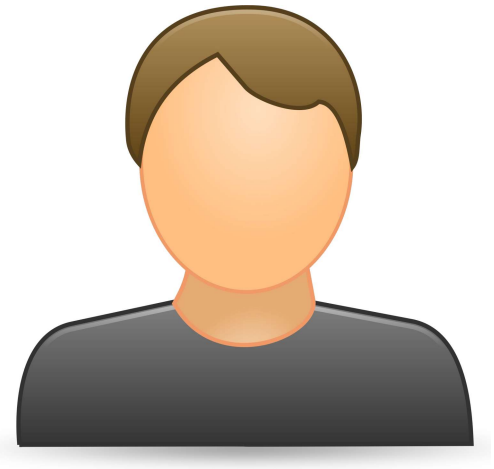


Database Users

- System Analysts
  - Business analysis and requirements gathering
- Database Designer
  - create database design (Conceptual Schema)
- Database Administrator (DBA)
  - Install DBMS.
  - Create Users and authorize access to db .
  - Maintain DB performance

Database Users Cont.

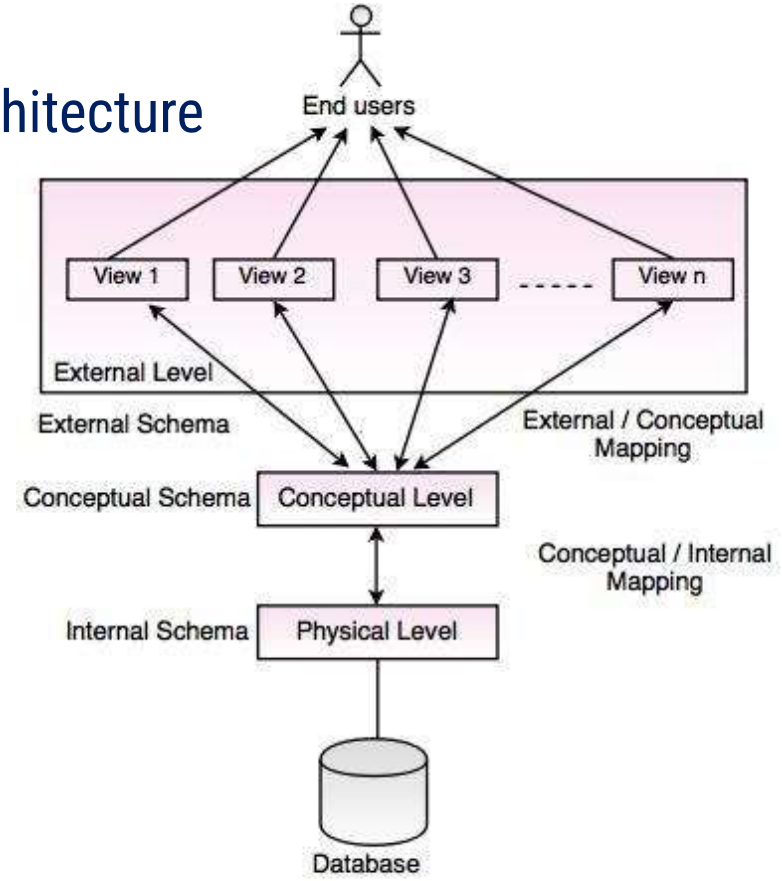
- Application programmers
  - Develop , test and debug the application .
- End users



Three Level/Schema Architecture

The three levels of database architecture

1. External Level
2. Conceptual Level
3. Physical Level



External Level

- External level is related to the data which is viewed by individual end users.
- This level includes a no. of user views or external schemas.
- This level is closest to the user.
- External view describes the segment of the database that is required for a particular user group and hides the rest of the database from that user group.

Conceptual Level

- Conceptual level describes the structure of the whole database for a group of users.
- It is also called as the data model.
- Conceptual schema is a representation of the entire content of the database.
- These schema contains all the information to build relevant external records.
- It hides the internal details of physical storage.

Physical Level

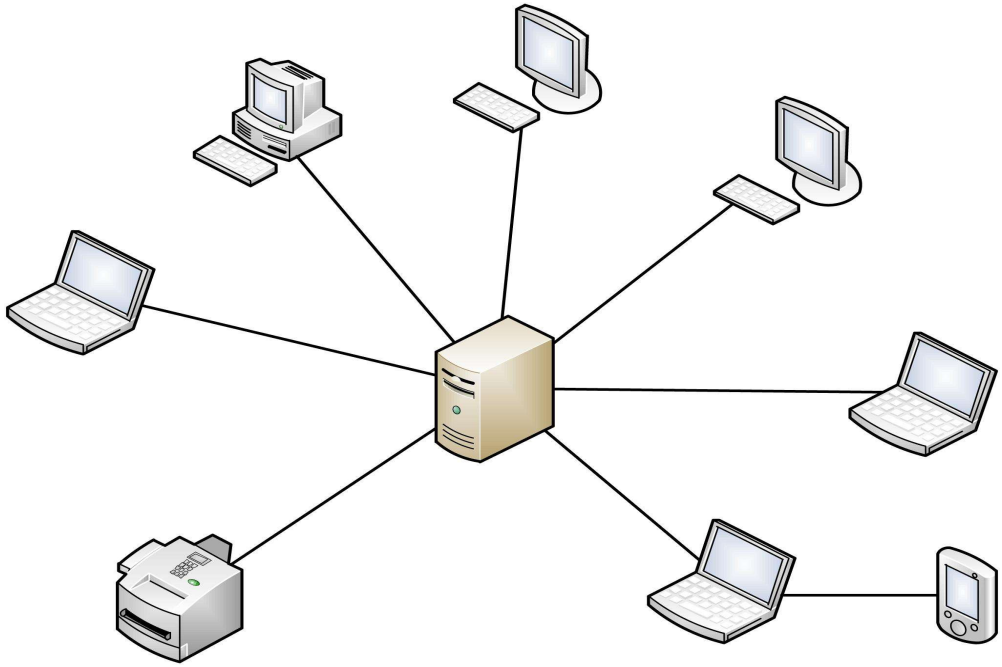
- Physical level describes the physical storage structure of data in database.
- It is also known as Internal Level.
- At lowest level, it is stored in the form of bits with the physical addresses on the secondary storage device.
- At highest level, it can be viewed in the form of files.
- The internal schema defines the various stored data types. It uses a physical data model.

Centralized Database Environment

- **Mainframe Environment**
  - Everything is on one machine and all processing made on one machine
  - The processing depends on server.

  - **Disadvantages**
    - The processing depends on one server
    - The performance is very slow
    - Database and application layer has single point of failure

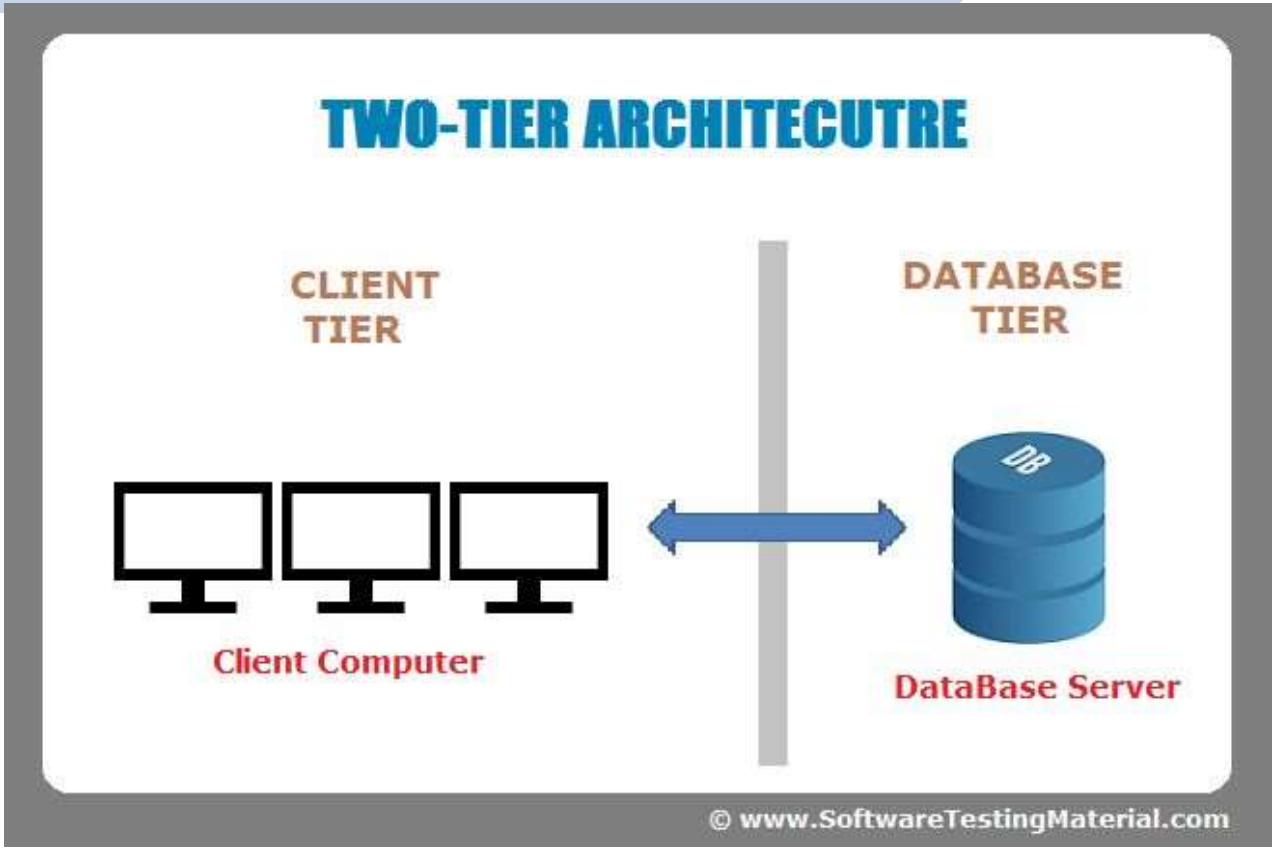
Mainframe Environment



Centralized Database Environment Cont..

- **Client/Server Environment**
  - Two-tier environment
    - The first tier is a database server .
    - The second tier is my client.Application is installed locally on every end-user machine.

Client / Server Environment



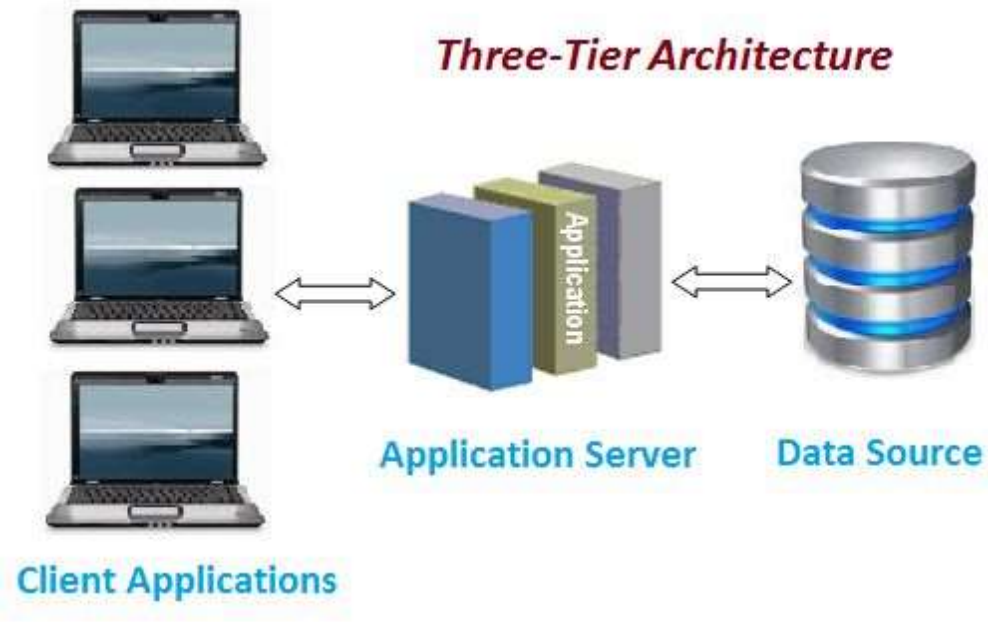
Client / Server Computing Environment

- **Advantages:**
  - Application layer isn't a single point of failure.

- **Disadvantages**
  - Database is a single point of failure.
  - High cost for support.

Centralized Database Environment Cont..

- **Internet Computing environment (Three-tier architecture)**
  - Three-tier environment
    - Thin Client.
    - Application Server.
    - Database Server.



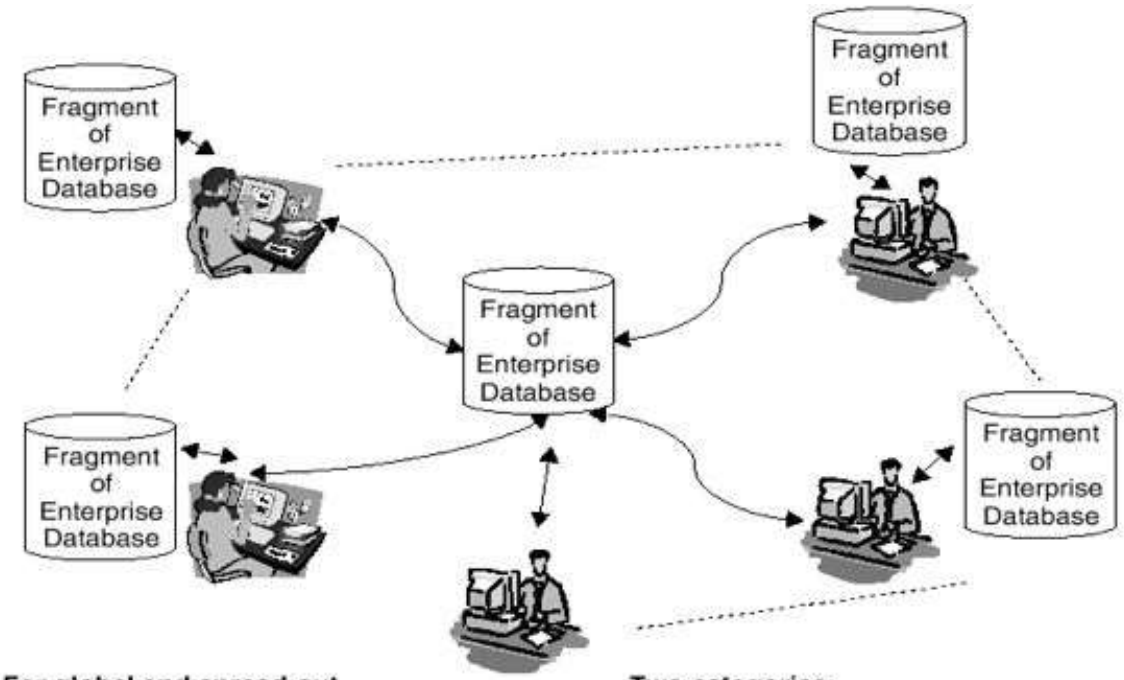
Internet Computing Environment

- **Advantages:**
  - lower cost for support and maintenance.
- **Disadvantages**
  - Application server is a single point of failure.
  - Database is a single point of failure.

Distributed Database

- Replication (Copy - Past)
  - Partial Replication
  - Full Replication
- Fragmentation (Cut - Past)

Distributed Database



For global and spread-out organizations, centralized databases not economical.

Enterprise data distributed across multiple computer systems.

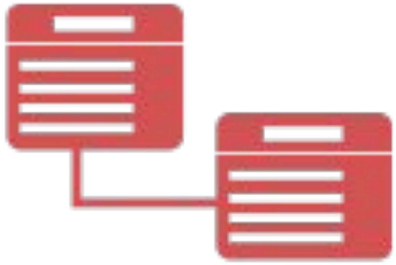
Two categories:

- **Homogeneous databases**
- **Heterogeneous databases**

ER Model

Entity Relationship Diagram

Concepts



ER Model

- **ER Model** is a high-level data model, developed by Chen in 1976. This model defines the data elements and relationships for a specified system. It is useful in developing a conceptual design for the database & is very simple and easy to design logical view of data.

Importance of ER Model

- ERD stands for Entity Relationship diagram.
- ER diagram shows the relationship between objects, places, people, events etc. within that system.
- ER Model is a graphical representation for designing the structure.
- Without ER diagrams you cannot make a database structure & write production code.
- It displays the clear picture of the database structure.

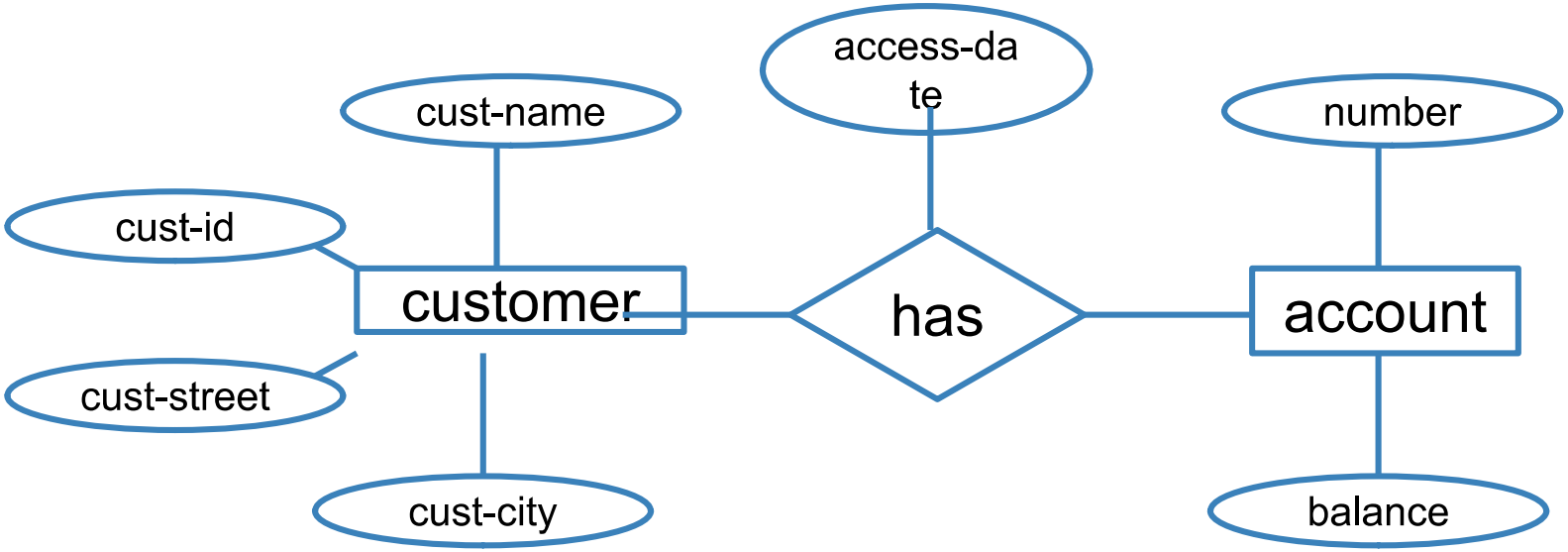
Basic constructs of the E-R model

- **Entities** - person, place, object, event, concept (often corresponds to a real time object that is distinguishable from any other object)

- **Attributes** - property or characteristic of an entity type (often corresponds to a field in a table)

- **Relationships** – link between entities (corresponds to primary key-foreign key equivalencies in related tables)

ER Diagram: Starting Example



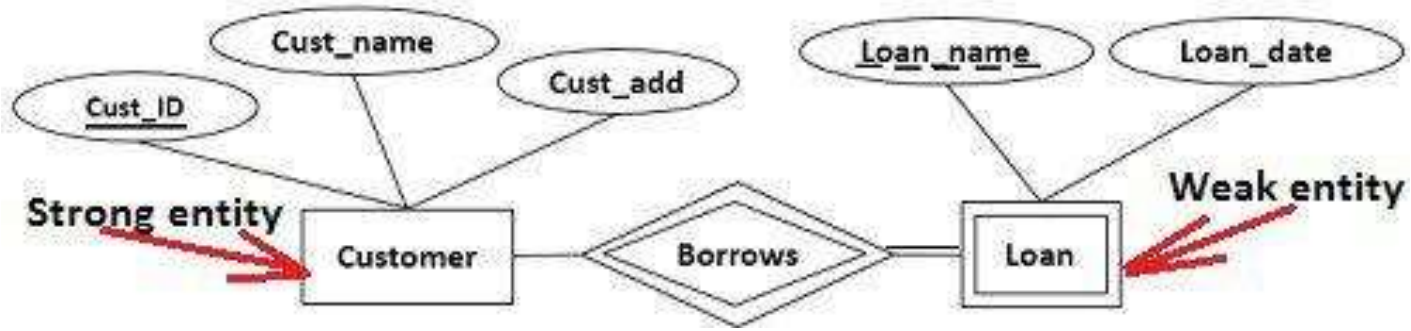
- **Rectangles**: entity sets
- **Diamonds**: relationship sets
- **Ellipses**: attributes

Entity

- A definable thing—such as a person, object, concept or event—that can have data stored about it.
- Think of entities as nouns.
- Examples: a customer, student, car or product.
- Typically shown as a rectangle.

Entity categories

- **strong entity** can be defined solely by its own attributes.
- **weak entity** cannot defined solely by its own attributes**,** An entity set that do not have sufficient attributes to form a primary key.



Entity keys

- Candidate Key
- Primary Key
- Foreign Key
- Composite Key
- Partial Key
- Alternate key
- Super Key

Primary Key

Identifier used to uniquely identify one particular instance of an entity.

- Can be one or more attributes.
- Must be unique .
- Value should not change over time.
- Must always have a value .

Entity keys

- **Candidate Key :**

when multiple possible identifiers exist, each is a candidate key.

- **Foreign Keys :**

Foreign keys reference a related table through the primary key of that related table.

- **Referential Integrity Constraint:**

For every value of a foreign key there is a primary key with that value in the referenced table e.g. if student name is to be used in a dormitory table then that name must exist in the student table.

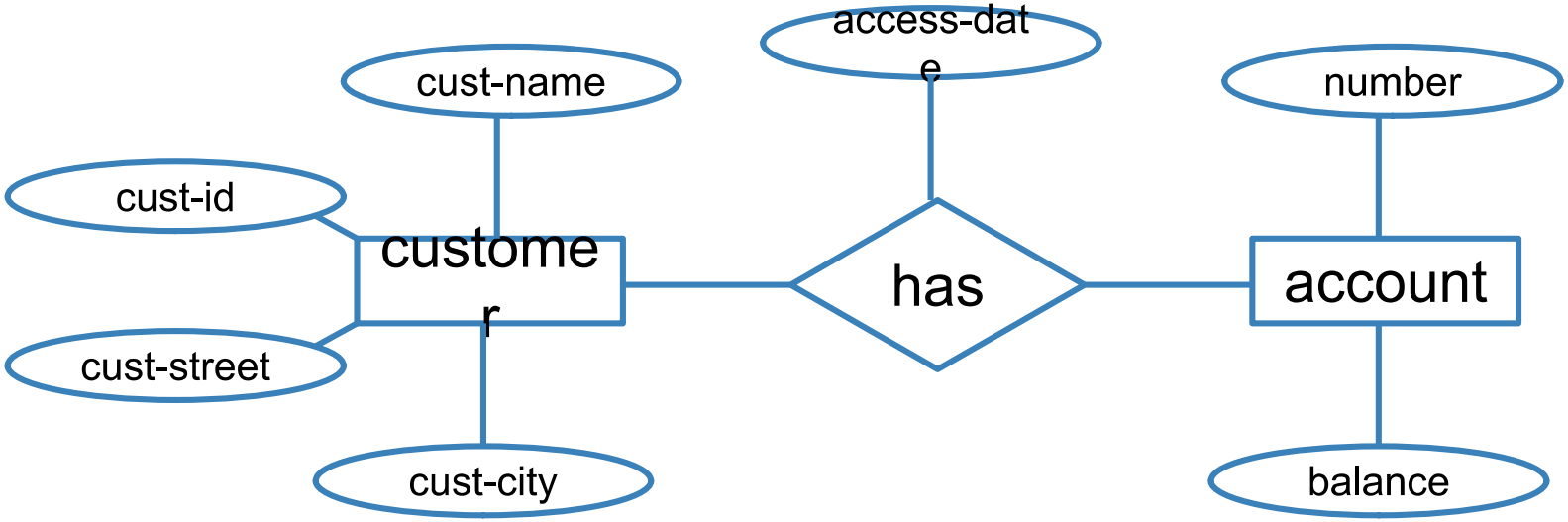
Attribute

- A property or characteristic of an entity. Often shown as an oval or circle.
- **Attribute categories:**

  - Simple (Scalars) Attribute
  - Composite Attribute
  - Derived Attribute
  - Multi-valued Attribute
  - Complex Attribute

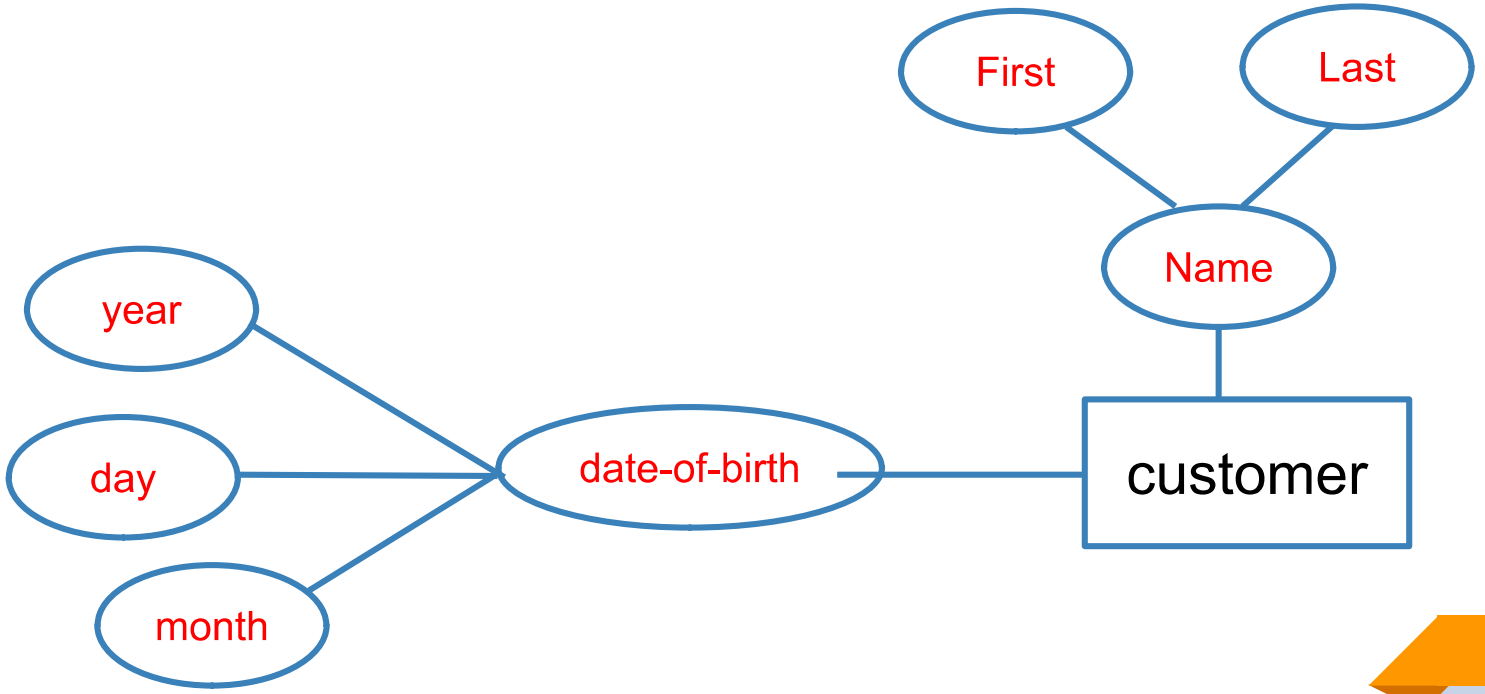
Simple Attribute

- Means the attribute value is atomic and can't be further divided, such as a phone number.



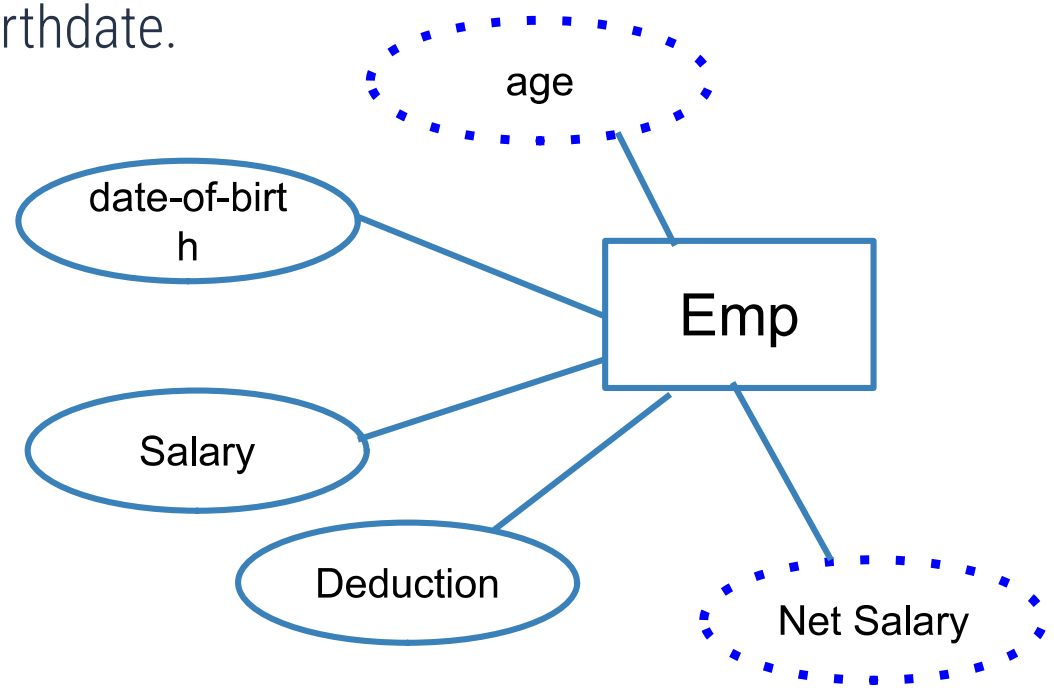
Composite Attribute

- Sub-attributes spring from an attribute



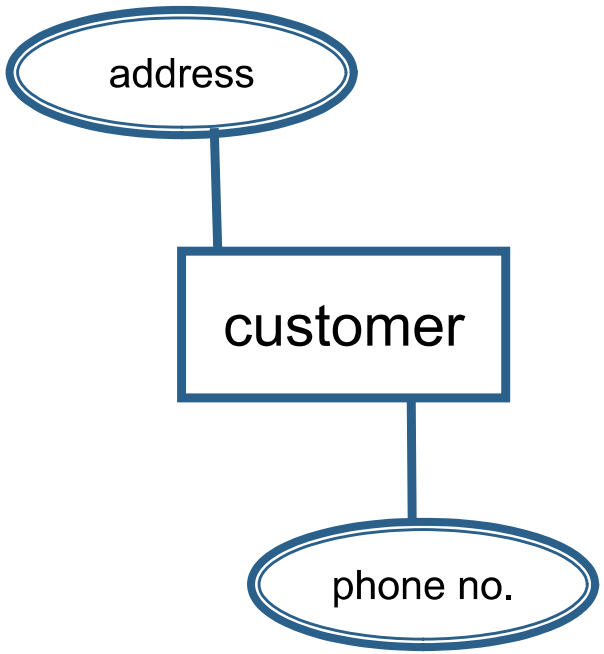
Derived Attribute

- Attributed is calculated or otherwise derived from another attribute, such as age from a birthdate.



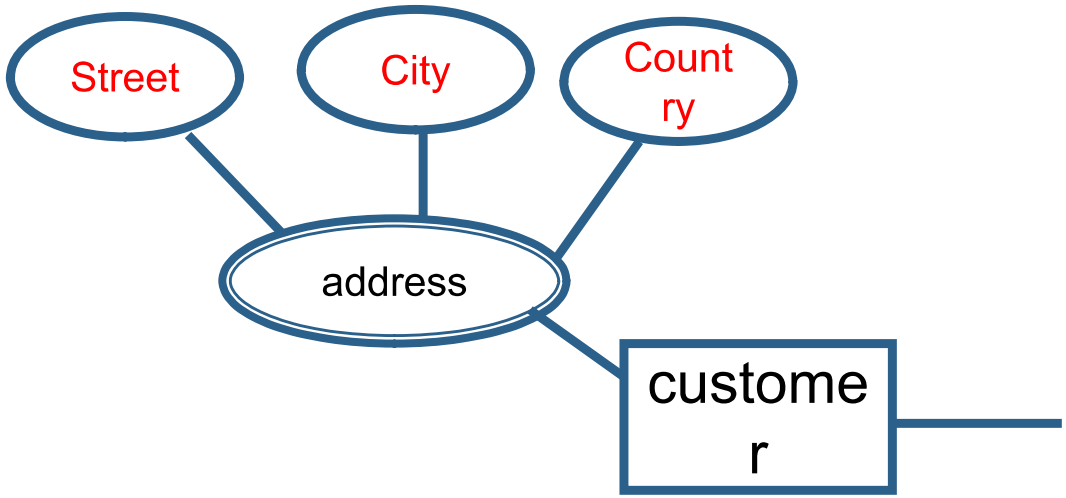
Multi-valued

- More than one attribute value is denoted, such as multiple phone numbers for a person.



Complex Attribute

- multi-valued + Composite



Relationship

- A Relationship is an association among several entities.
- A relationship may also have attributes
- For example, consider the entity sets customer and loan and the relationship set borrower. We could associate the attribute date-issued to that relationship to specify the date when the loan was issued.

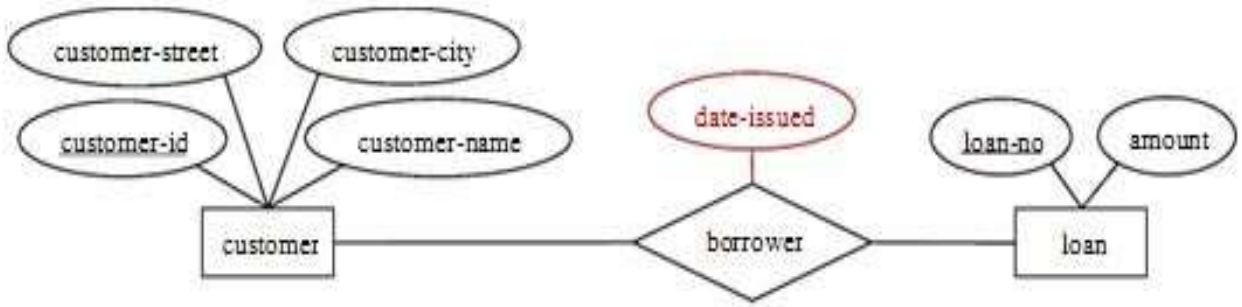


Figure: Descriptive attribute *date-issued*.

Relation Properties

Relation has three Properties:

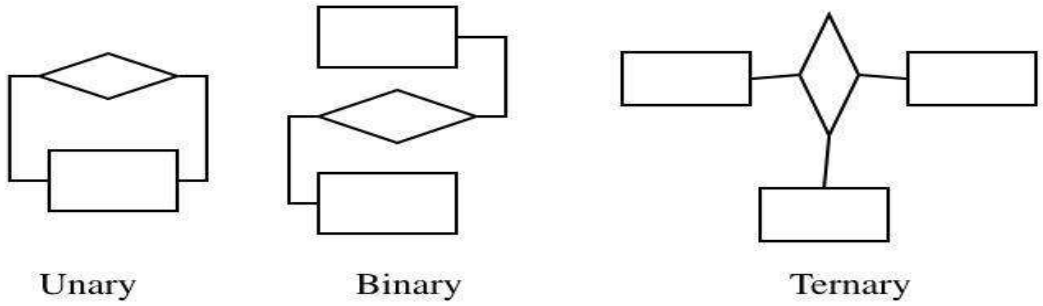
- Degree of Relationships
- Cardinality Constraint
- Participation Constraint

Degree of Relationships

- number of entity types that participate in a relationship

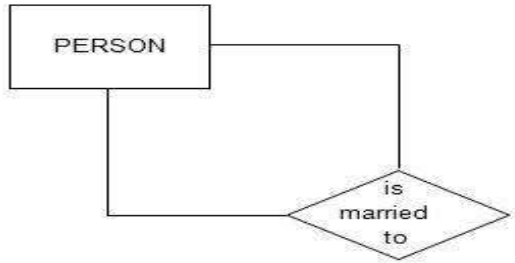
Three cases:

- **Unary**: between two instances of one entity type
- **Binary**: between the instances of two entity types
- **Ternary**: among the instances of three entity types



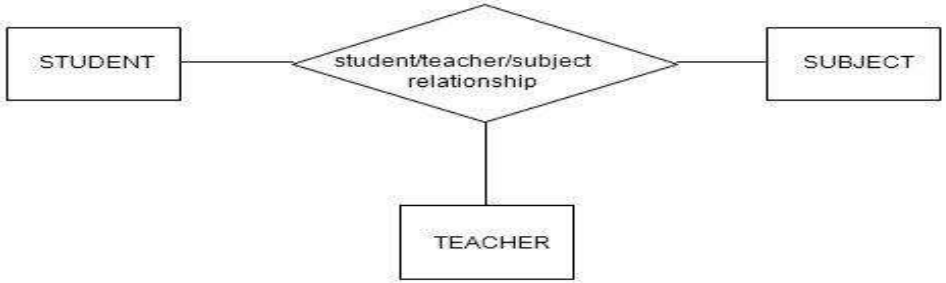
Degree of Relationships

Unary



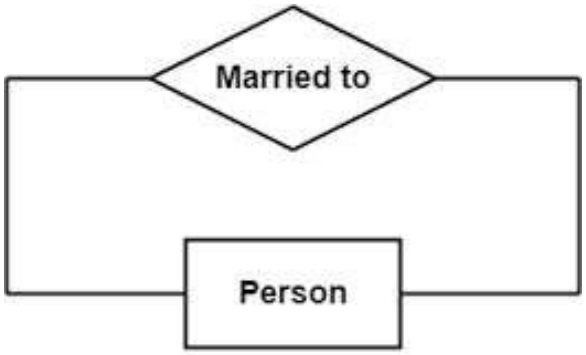
Binary

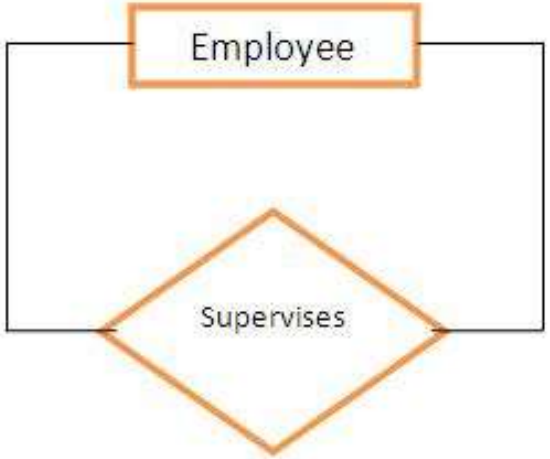
Ternary



Recursive Relationship (Unary)

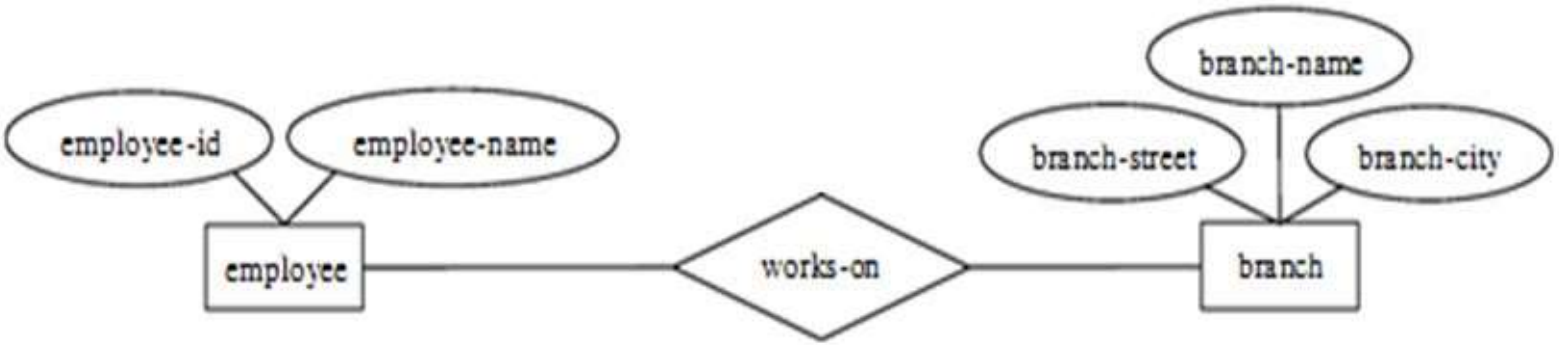
- **Recursive Relationships -** A relationship in which the same entity participates more than once.





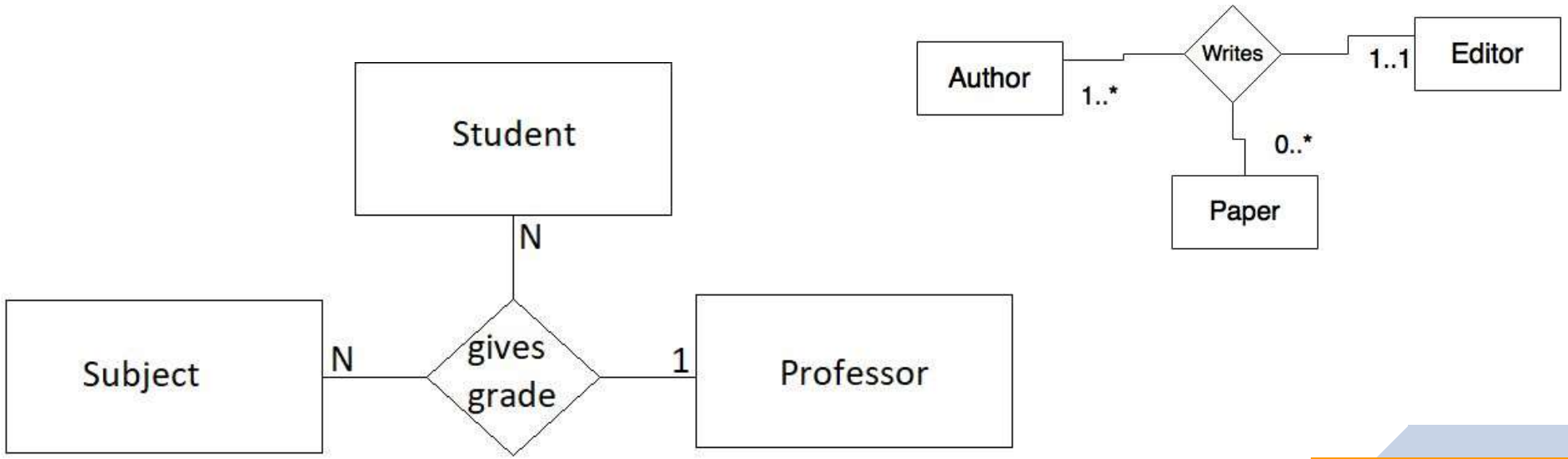
Binary Relationship

- **A binary relationship** set is of degree 2.



Ternary Relationship

- **Ternary relationship** set is of degree 3.



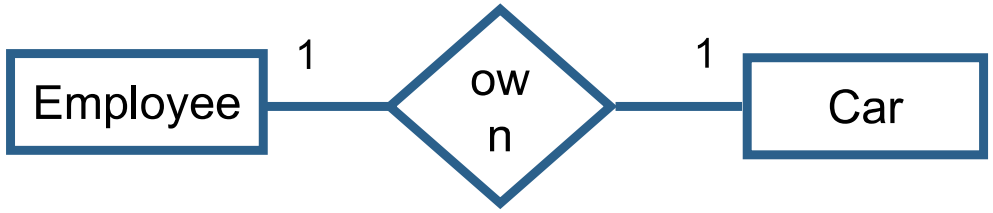
Cardinality

How many instances of one entity will or must be connected to a single instance from the other entities.

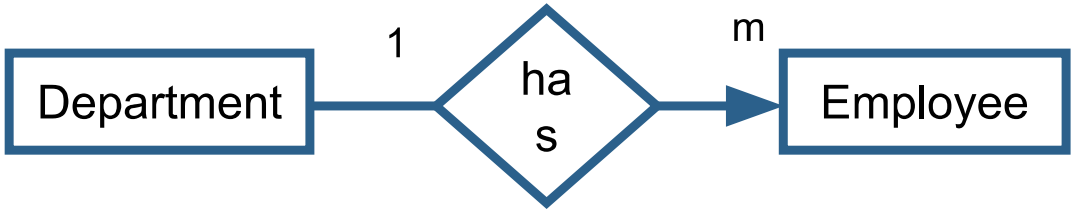
- One-One Relationship
- One-Many Relationship
- Many- Many Relationship

Mapping Cardinalities

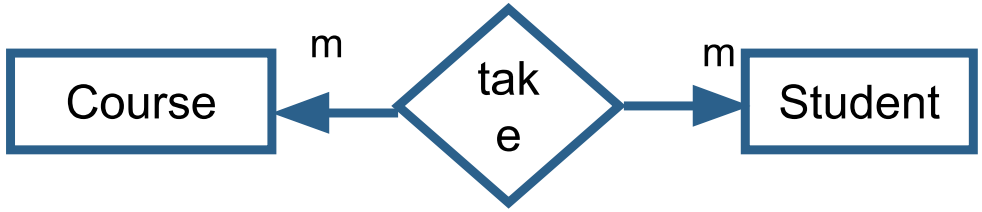
- One-to-One



- One-to-Many



- Many-to-Many



PARTICIPATION CONSTRAINT

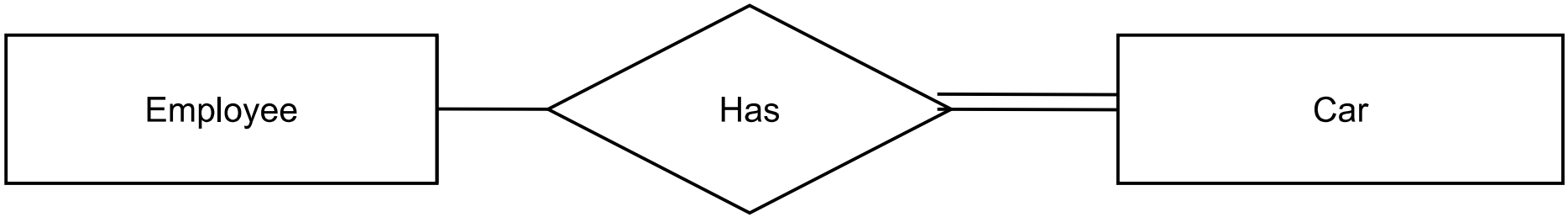
- An employee MUST work for a department

An employee entity can exist only if it participates in a WORKS_FOR relationship instance
So this participation is **TOTAL**

- Only some employees manage departments

The participation is **PARTIAL**

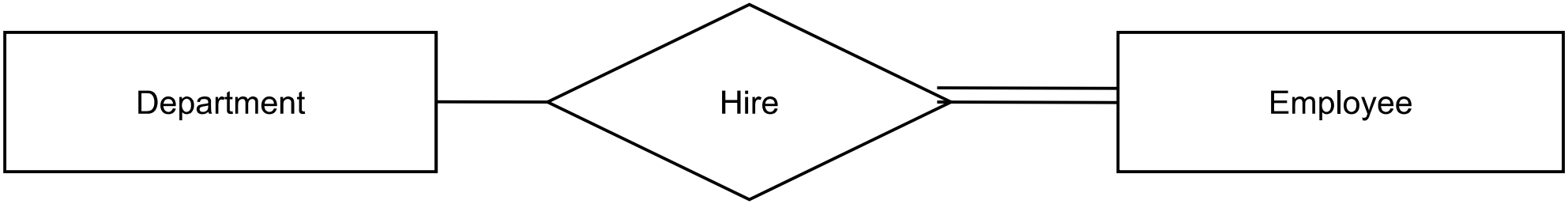
PARTICIPATION CONSTRAINT



-An Employee may have a car.

-A Car must be assigned to particular employee

PARTICIPATION CONSTRAINT

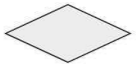
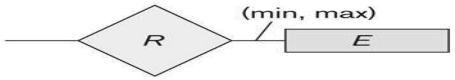


- A department may hire many employees (Zero or more)

- An employee must be employed by a department

(Department membership is Optional, Employee membership is Mandatory)

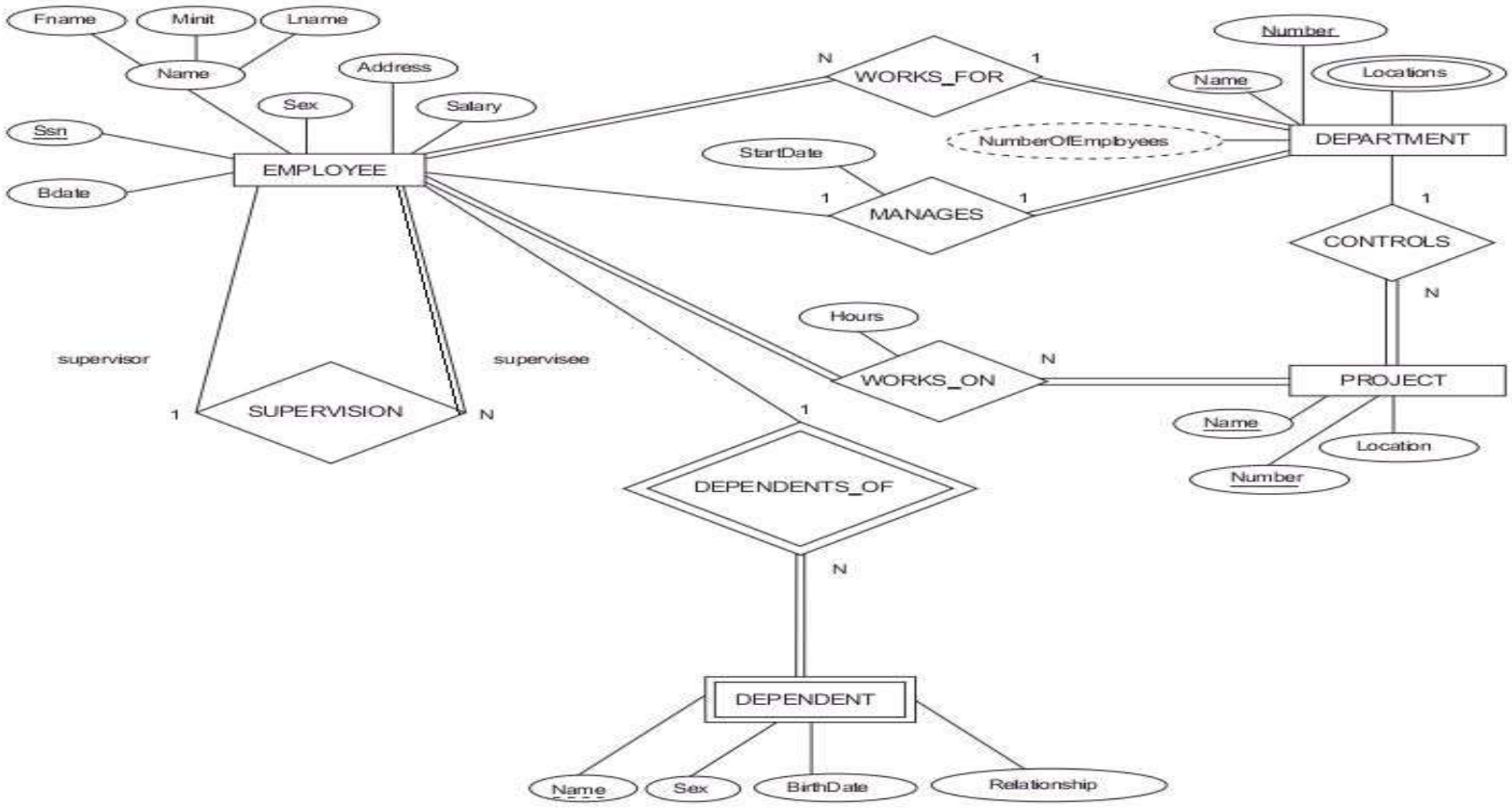
Summary of notation for ERD

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Indentifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
	Total Participation of E_2 in R
	Cardinality Ratio 1 : N for E_1:E_2 in R
	Structural Constraint (min, max) on Participation of E in R

Case Study

Company

Result


Fname
Minit
Lname
Name
Address
Sex
Salary
Ssn
Bdate
EMPLOYEE
N
WORKS_FOR
1
Number
Name
Locations
DEPARTMENT
StartDate
NumberOfEmployees
1
MANAGES
1
1
CONTROLS
N
Hours
supervisor
supervisee
WORKS_ON
N
PROJECT
1
SUPERVISION
N
Name
Location
Number
1
DEPENDENTS_OF
N
DEPENDENT
Name
Sex
BirthDate
Relationship

THANKS!

Any questions?